9.1 Using Normal Distributions

Name _____ Period ____ Date _____

Algebra II 9.1 (30-minute plan) · Student sheet

I can…

- use the 68-95-99.7 rule to find areas under a normal curve.
- find probabilities for normally distributed data.
- tell whether data are approximately normal and how skew affects the mean and median.

Vocabulary · fill in the missing word

Normal distribution — a _____, symmetric distribution with mean μ and standard deviation σ

Normal curve — the bell-shaped graph of a normal _____; total area 1

68-95-99.7 rule — about 68%, 95%, and 99.7% of the data are within 1, 2, and 3 standard _____ of the mean

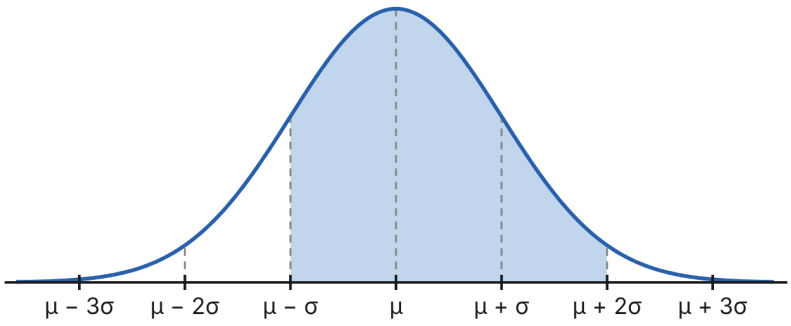
Skewed — not _____: the mean is pulled toward the tail

Part 1 · Notes and examples (fill in each note, then work the example)

The 68-95-99.7 rule: Within 1σ of the mean: _____% within 2σ: _____% within 3σ: _____%

Example 1

What percent of the area under the normal curve is shaded?

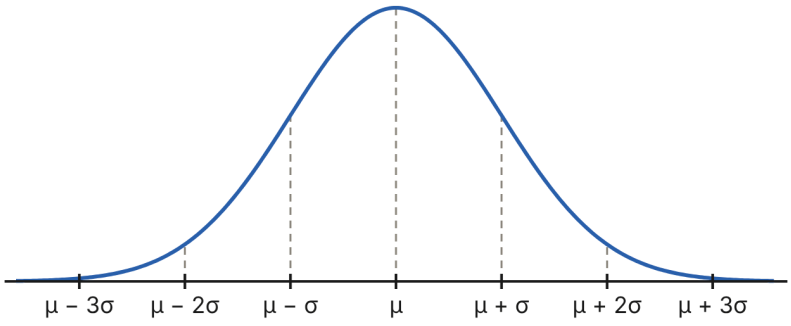


Areas of the bands: Bands on each side: μ to σ: _____%, σ to 2σ: _____%, 2σ to 3σ: _____%, beyond 3σ: _____%

Example 2

A normal distribution has mean μ and standard deviation σ. Find each probability.

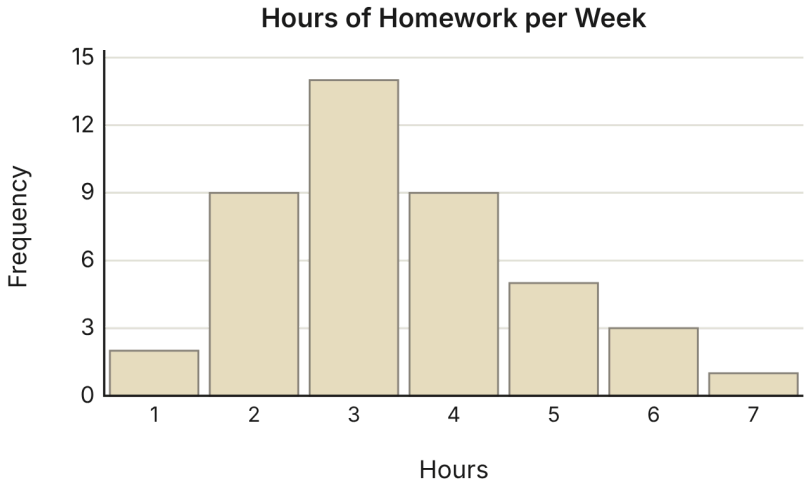
a. $P(x \le \mu - \sigma)$ b. $P(x \ge \mu + 2\sigma)$



Skewed data: Skewed right: mean _____ median. Skewed left: mean _____ median.

Example 3

Does the histogram show data that are approximately normal? Is the mean greater or less than the median?



Part 2 · Partner practice

Partner practice

A normal distribution has a mean of 150 and a standard deviation of 10. Find P(x < 130).